PRACTICAL 11 – ROUTING AT THE NETWORK LAYER

AIM

To configure and verify static routing and RIP routing between multiple networks using Cisco Packet Tracer.

PROCEDURE

1. Place three routers and the required PCs in the Cisco Packet Tracer workspace.
2. Connect the devices using suitable Ethernet and serial cables.
3. Assign IP addresses and subnet masks to router interfaces and PCs. Configure the default gateway on each PC.
4. Enable router interfaces with the `no shutdown` command. If a serial interface is the DCE end, configure an appropriate clock rate.
5. Configure static routes using the `ip route` command. Add a backup route with a higher administrative distance if a redundant path is required.
6. Verify configured static routes with `show ip route static`.
7. Configure RIP version 2 on the routers and advertise the required networks using the `network` command.
8. Use `ping` and `tracert` from the PCs to test end-to-end connectivity and inspect the route taken.

SAMPLE STATIC ROUTE CONFIGURATION

Use the following command format, replacing the placeholders with the appropriate values:

```
ip route [destination-network] [subnet-mask] [next-hop-IP]
```

SAMPLE RIP CONFIGURATION

Enter the following commands in router configuration mode:

```
router rip
version 2
network [network-address]
no auto-summary
```

INPUT

Network addresses, subnet masks, next-hop IP addresses, and router-interface settings are used to configure the routing protocols.

OUTPUT

The routing tables show the configured static routes and RIP-learned routes. Successful ping responses and traceroute results indicate connectivity and help identify the path between networks.

RESULT

Static routing and RIP routing were configured and tested in Cisco Packet Tracer. Connectivity between the PCs was verified using ping and tracert.